

EV Power - Lab 4 Project Report

Part 0: libraries

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0
—
✓ dplyr      1.1.4      ✓ readr      2.1.5
✓ forcats    1.0.1      ✓ stringr    1.5.2
✓ ggplot2    4.0.0      ✓ tibble     3.3.0
✓ lubridate  1.9.4      ✓ tidyr      1.3.1
✓ purrr      1.1.0
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(leaflet)

setwd("/Users/briannajordan/stat133/ev-power-briannajordan")

renew_2021 <- read.csv("data/renew-use-2021.csv")
renew_2022 <- read.csv("data/renew-use-2022.csv")
renew_2023 <- read.csv("data/renew-use-2023.csv")
total_energy_2021 <- read.csv("data/total-use-2021.csv")
total_energy_2022 <- read.csv("data/total-use-2022.csv")
total_energy_2023 <- read.csv("data/total-use-2023.csv")
ev_registrations_2023 <- read.csv("data/ev-registrations-by-state-2023.csv")
avg_energy_price_2021_2023 <- read.csv("data/av-energy-price-2021-2023.csv")
```

Part 1: Defining Research Question

Chosen Question: How has the share of renewable energy changed from 2021 to 2023 across the US?

Part 2: Data Preparation and Cleaning

```
clean_df <- function(df) {df |>
  rename_with(tolower) |>
  rename(state = 1) |>
  mutate(state = str_to_title(state))}

renew_2021 <- clean_df(renew_2021)
renew_2022 <- clean_df(renew_2022)
renew_2023 <- clean_df(renew_2023)
total_2021 <- clean_df(total_energy_2021)
total_2022 <- clean_df(total_energy_2022)
total_2023 <- clean_df(total_energy_2023)

head(renew_2021)
```

	state	energy_source	renewable_use_2021
1	Ak	Biomass	≈3153
2	Ak	Geothermal	186 MMBtu
3	Ak	Hydropower	5763 about
4	Ak	Solar Energy	~45
5	Ak	Wind Energy	451 USD
6	Al	Biomass	198543 est.

Part 3: Joining / Pivoting Datasets for Analysis

```
clean_df <- function(df) {df |>
  rename_with(tolower) |>
  rename(state = 1) |>
  mutate(state = str_to_title(state))}

renew_2021 <- clean_df(renew_2021)
renew_2022 <- clean_df(renew_2022)
renew_2023 <- clean_df(renew_2023)
total_2021 <- clean_df(total_energy_2021)
total_2022 <- clean_df(total_energy_2022)
total_2023 <- clean_df(total_energy_2023)

sum_renew <- function(df) {
  df |>
  rename(renewable_use = starts_with("renewable_use")) |>
  mutate(renewable_use = as.numeric(renewable_use)) |>
  group_by(state) |>
  summarize(renewable_use = sum(renewable_use, na.rm = TRUE))}

renew_2021_clean <- sum_renew(renew_2021)
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `renewable_use = as.numeric(renewable_use)`.
Caused by warning:
! NAs introduced by coercion
```

```
renew_2022_clean <- sum_renew(renew_2022)
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `renewable_use = as.numeric(renewable_use)`.
Caused by warning:
! NAs introduced by coercion
```

```
renew_2023_clean <- sum_renew(renew_2023)
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `renewable_use = as.numeric(renewable_use)`.
Caused by warning:
! NAs introduced by coercion
```

```
tidy_total <- function(df) {df |>
  select(-state) |>
  pivot_longer(everything(), names_to = "state", values_to = "total_use") |
  >
  mutate(total_use = as.numeric(total_use),
    state = str_to_title(state))}

total_2021_tidy <- tidy_total(total_2021)
total_2022_tidy <- tidy_total(total_2022)
total_2023_tidy <- tidy_total(total_2023)

combine_year <- function(renew, total, year_val) {
  left_join(renew, total, by = "state") |>
  mutate(
    total_use = as.numeric(total_use),
    renewable_use = as.numeric(renewable_use),
    year = year_val,
    renew_share = ifelse(is.finite(total_use) & total_use > 0,
      (renewable_use / total_use) * 100,
      NA_real_) |>
  select(state, year, renew_share)}

all_years <- bind_rows(
  combine_year(renew_2021_clean, total_2021_tidy, 2021),
  combine_year(renew_2022_clean, total_2022_tidy, 2022),
```

```

combine_year(renew_2023_clean, total_2023_tidy, 2023))

all_years_clean <- all_years |>
  group_by(state, year) |>
  summarize(
    renew_share = mean(renew_share, na.rm = TRUE),
    .groups = "drop")

head(all_years_clean)

```

```

# A tibble: 6 × 3
  state year renew_share
<chr> <dbl>      <dbl>
1 Ak    2021         0
2 Ak    2022         0
3 Ak    2023        26.6
4 Al    2021         9.90
5 Al    2022         9.03
6 Al    2023         9.35

```

Part 4: Mapping Visualization

```

all_years_clean2 <- all_years_clean |>
  mutate(
    state = toupper(state),
    state = tolower(state.name[match(state, state.abb)])
  ) |>
  filter(!is.na(state))

us_states <- st_as_sf(maps::map("state", plot = FALSE, fill = TRUE)) |>
  mutate(state = tolower(ID))

plot_renew_map <- function(year_val) {
  df <- all_years_clean2 |>
    filter(year == year_val)

  map_df <- us_states |>
    left_join(df, by = "state")

  ggplot(map_df) +
    geom_sf(aes(fill = renew_share), color = "white") +
    scale_fill_gradient(
      name = "Renewable Share (%)",
      low = "lightblue",
      high = "darkgreen",
      na.value = "gray") +

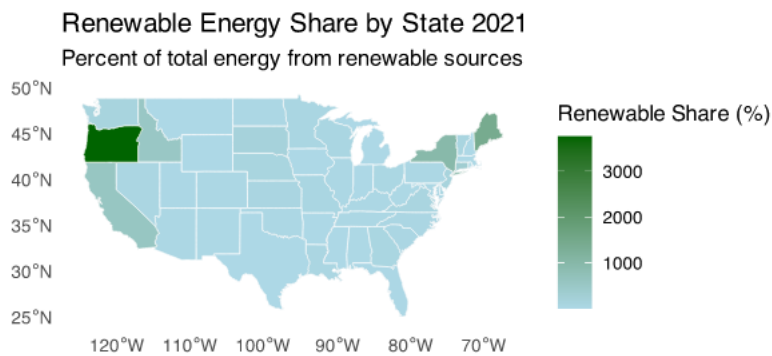
```

```

labs(title = paste("Renewable Energy Share by State", year_val),
      subtitle = "Percent of total energy from renewable sources") +
theme_minimal() +
theme(legend.position = "right", panel.grid.major = element_blank())}

plot_renew_map(2021)

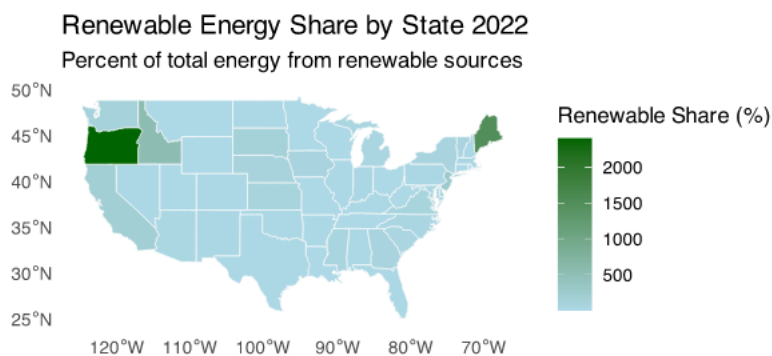
```



```

plot_renew_map(2022)

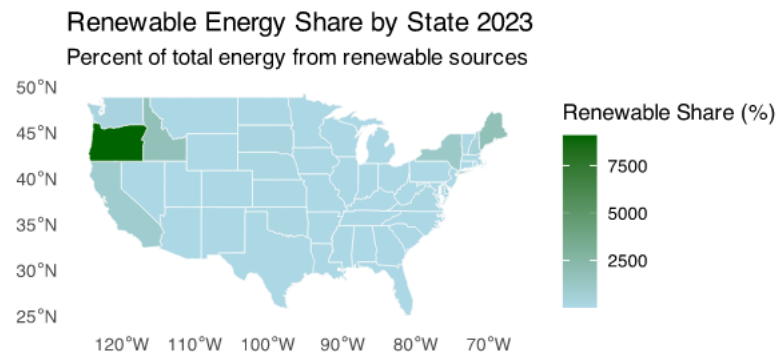
```



```

plot_renew_map(2023)

```



Analysis

From 2021 to 2023, renewable shares have stayed relatively stable with greater amounts seen in the west coast and the east coast. Oregon has the largest percentage of renewable energy because it is very dark green. In terms of our main research question, although EVs decrease emissions, my maps show that the electricity powering them varies significantly across the U.S. In states with high renewable energy shares (e.g., California, Oregon, Vermont), EVs are low-carbon. In states with lower renewable shares, EV charging still relies heavily on fossil fuels, reducing but not eliminating emissions.